CSCI 141: Modern Programming Fundamentals
Fall 2026 Syllabus

About this course

An introduction to computer programming and computational problem solving. Students will learn to create algorithmic solutions to problems and to express their solutions as computer programs. Programming assignments will focus on problems taken from various disciplines including computational and data sciences.

It does not assume any prior programming experience. The course will use the Python programming language as well as many Python data science tools via the Anaconda software stack. Throughout the course, we will also use modern computational tools, including integrated development environments (IDEs) and evolving artificial intelligence (AI) tools, with an emphasis on using these tools responsibly and critically to support programming, testing, debugging, and problem solving.

Topics include:

- Variables and data types
- Functions
- Control flow
- Iteration with loops
- Built-in data structures, including lists, tuples, dictionaries, etc.
- File input/output
- Data structures from packages such as pandas and numpy
- Obtaining, cleaning, and exploring data
- Testing and debugging
- Object-oriented programming

We will use the Python programming language as a tool for the manipulation, analysis, and visualization of a breadth of data types: spatial, temporal, numerical, and textual. This course explores the writing and manipulation of data in different formats, and standardizing data into canonical formats using Python tools such as the "Pandas" data analysis toolkit. Visualizations will exploit the use of Python libraries such as matplotlib and seaborn. We will draw examples from a variety of fields including economics, statistics, linguistics, and other natural and social sciences.

Corequisite: CSCI 141L

This is a 4 credit course which also includes a weekly lab session held by the TAs, who are Computer Science graduate students. **ATTENDANCE AT LAB SESSIONS IS MANDATORY.** Throughout the lab, don't hesitate to raise questions and discuss any uncertainties related to the course materials and assignments.

Instructor and Meeting Information

Mei Zhang

mzhang11@wm.edu

Office Hours: To be determined (will be announced soon)

	Course ID	Days and Time	Place
Section 01	CSCI 141 - 01	MWF:1000-1050	Blow Memorial Hall 331
Section 02	CSCI 141 - 02	MWF:1100-1150	Blow Memorial Hall 331
Section 03	CSCI 141 - 03	MWF:1300-1350	Integrated Science Center 3349

Labs: Integrated Science Center 2358

Location note: If you are new to campus, Integrated Science Center 2358 and Integrated Science Center 3349 are both located in ISC4 (Integrated Science Center IV). Although this page (https://events.wm.edu/event/view/computerscience/372921) is for a past event, it provides helpful information for locating the building.

Required Materials

- We will use **Blackboard** for course organization, and assignment submissions will be accepted through Blackboard. Other assessments may also be administered through Blackboard.
- We will also use **Gradescope** for assignment submissions, the link for which will be available on Blackboard.
- Programming in this course will be done in **Python3** using the Anaconda Python Stack (link is in Setup document in Module 0). We will provide

instructions and assistance for setting up Anaconda for Windows 7/10, Mac OS X, or Linux on your personal machine. The environment will also be available on Department Linux systems in McGl 121. **You must not use Python 2.7, which is often the default on personal computers**.

- We will use **Piazza** (www.piazza.com) for class discussion and questions. Different students often pose similar questions, and answering questions in a public forum benefits more people, including those who might not have thought to ask the question. You will receive an e-mail invitation to Piazza; please accept it. Piazza has changed their model to one that asks for voluntary contributions. This is a function of Piazza and is not under the control of the instructors. You are not asked by the instructors nor are required by this course to contribute to Piazza.
- It is your responsibility to **bring an operable device to every class** in order to participate in lecture activity. You will need to access Jupyter notebooks during class. Please note that students are expected to be respectful of the class, and refrain from disrupting the class in any way. Any sort of audio/video recording or picture-taking is forbidden during class.
- We will use Poll Everywhere ("PollEv") throughout this semester. With PollEv, you use your computer, tablet, or phone to answer questions, take a poll, discuss, and more.

Textbook

There is no required textbook for this course. For those of you who like to have a reference, you can consult optional textbooks:

Think Python: How to Think Like a Computer Scientist | Edition: 2
Author: Allen B. Downey, ISBN: 9781491939369. The library has an online version that you can access for free at: https://proquest-safaribooksonline-com.proxy.wm.edu/book/programming/python/9781449332006/firstchapter

Python Crash Course, 3rd Edition: A Hands-On, Project-Based Introduction to Programming Eric Matthes. Publisher: No Starch Press; 3rd edition (January 10, 2023) ISBN-10 : 1718502702; ISBN-13 : 978-1718502703

You may also want to consult the following textbook for Module 4 of the course:

Python for data analysis / Wes McKinney. Sebastopol, Calif. : O'Reilly ©2013
The library has an online version that you can access for free at:
http://proquest.safaribooksonline.com.proxy.wm.edu/9781449323592

Important Dates

September 4, 2026: Add/drop deadline

October 26, 2026: Last day for course withdrawal

December 10, 2026 at 10:00 PM: Final exam deadline

Grade Computation

Your grade in this course will be computed as follows:

Lab work (CSCI 141L)		10%
Comprehension Quizzes (Exclude the Course Syllabus Agreement Quiz and Final Exam Guidelines Quiz)		10%
In-term Problem Sets		16%
In-term Projects		28%
Midterm		8%
Final Exam	Part 1 (Final Project)	20%
	Part 2 (Final Problem Set)	8%

Your final grade for the semester will be determined using the following scale. If S is your total score for the semester, then

$S < 60\%$	F				
$60\% \leq S < 63\%$	D-	$63\% \leq S < 67\%$	D	$67\% \leq S < 70\%$	D+
$70\% \leq S < 73\%$	C-	$73\% \leq S < 77\%$	C	$77\% \leq S < 80\%$	C+
$80\% \leq S < 83\%$	B-	$83\% \leq S < 87\%$	B	$87\% \leq S < 90\%$	B+
		$90\% \leq S < 94\%$	A-	$94\% \leq S \leq 100\%$	A

If appropriate, we will curve those standards in your favor, but we will not raise them. Note that borderline grades which are more than 0.25% from the next grade level may not be rounded up. We reserve the right to consider participation in borderline grade decisions.

Deadlines are strictly enforced in this course. Your submission will only be available for credit if submitted prior to the deadline.

Comprehension Quizzes

There are Comprehension Quizzes for each section of the course. These quizzes are designed to assess your understanding of core concepts and ideas from the readings, lectures and other additional materials you may receive.

These quizzes can be completed as many times as desired until the deadline. Once the deadline has passed, the quizzes will **no longer be available for credit**, even though you may see the score showing up on your blackboard page because of the automated scoring system. Only your last submitted attempt by the deadline will contribute towards your final grade in the course.

In-term Problem Sets

There will be four **equally weighted** in-term problem sets in this course, which are administered via Blackboard. They will be given over a week in advance. You may drop your lowest in-term problem set score. If you miss an in-term problem set for any reason, that will be your drop. Your drop also covers technical errors.

Do not expect to be able to make up problem sets once the deadline has passed. Once the deadline has passed, the problem set will **no longer be available for credit**, even though you may see the score showing up on your blackboard page because of the automated scoring system.

In-term Projects

There will be four **equally weighted** in-term programming projects, one each for Modules 1, 2, 3, and 4. The projects give you a chance to demonstrate what you have learned in each module, as well as the previous modules, as programming is a cumulative skill. Each project will showcase code that you write on your own and some may also include a debugging task, which helps build your skills in debugging so that you can debug your own code more effectively. Finally, you will have the chance to reflect on your work for each project and will share your thoughts, struggles, and plans for future work with us via a .pdf reflection.

Each and every one of you has the ability to take a 48-hour extension on any TWO separate projects of your choice (excluding the final project). This means that for two of the in-term projects, you may choose to extend the deadline by 48 hours. An extension only applies to a single project, e.g., if you use an extension on Project 2, it does not also cover Project 3.

If you have used all of your extensions, projects may be submitted within 48 hours of the original deadline for up to 50% credit.

Projects received more than 48 hours after the original deadline, or not turned in at all will be given a grade of 0.

The extensions cover any and all cases for which you would request an extension, including but not limited to personal issues, medical issues, academic issues, and technological issues. Projects will be given over a week in advance, and choosing to start the project a day or two before the deadline does not warrant a longer extension.

Midterm Exam

The midterm exam will be administered in a traditional in-person format. It will be a **closed-book, closed-device** exam, completed with pen or pencil on paper. I will post an announcement with more details as we get closer to the exam date.

Final Exam

The Final Exam consists of two parts:
Part 1: Final Project,
Part 2: Final Problem Set.

The time and date of the Final Exam is determined by the Office of the University Registrar, and any changes to exam deadlines must be granted by the Office of Undergraduate Academic Affairs or the Office of Academic Wellbeing in advance of the due date. The deadline for the final exam submission is: **Thursday, December 10, 2026 at 2200 (10:00 PM) EST.** We cannot accept ANY assignments after Thursday, December 10, 2026 at 2200 (10:00 PM) EST.

Late submissions cannot be accepted because the instructors do not have the authority to change final exam deadlines. If no submission is received by the deadline, the final exam score will be 0. Extensions can only be granted by the Office of Undergraduate Academic Affairs or the Office of Academic Wellbeing.

You must complete the Final Exam Guidelines Quiz with a score of 100% before you submit the Final Exam. If you fail to complete the Final Exam Guidelines Quiz with a score of 100% before 10:00 PM EST on Thursday, December 10, 2026, there will be significant grade penalties on the Final Exam.

(1) Part 1: Final Project

The final project comprises 20% of your grade in the course. It will incorporate elements from all of the course material. The final project will be given over a week in advance.

For all Projects, including in-term projects and final projects:

- **It is your responsibility to ensure that you have submitted gradable files and the correct version of your project.** If your files are not gradable, you will receive a 0. The last version of the project you submit by the deadline will be graded, even if you submitted it in error. You must submit all files in your last submission; we will not collate files from multiple submissions.
- You will submit your projects via Gradescope to an auto-grader. Projects are submitted as .py files, NOT .ipynb files. Your write-up must be a .pdf. If you submit a .docx or any other file, Gradescope will not render it, and you will get a 0 for the write-up.
- Failure to check your files prior to your final submission does not warrant another opportunity to submit.
- After you submit your work on the Gradescope, you should receive an automated email from Gradescope that includes the day and time you submitted the work, the assignment Due Date, the Late Due Date, and a link to your submission. That email is the **ONLY proof of your submission**, and you should save that email. If, for any reason, we didn't receive your submission on Gradescope, it will be considered a **non-submission** unless you can show the proof.
- Emailed submissions will not be accepted.

(2) Part 2: Final Problem Set

The final problem set comprises 8% of your grade in the course and will incorporate elements from all of the course material. The final problem set is administered via Blackboard and will be available over a **48-hour window**, which means it will be given 48-hour in advance.

Please note that when the final problem set is available, you may not be able to view the in-term problem set submissions.

For all Problem Sets, including in-term problem sets and final problem sets:

- You will receive **partial scores** if you select only some of the correct answers. You will not receive total scores if you choose all solutions (where some answers are incorrect). Therefore, we need to turn on negative scores - you will get **Negative Points for Incorrect Answers**. The partial credit percentage for each question may be different, depending on the difficulty level.
- If the problem set allows multiple attempts, only your last submitted attempt by the deadline will contribute towards your final grade in the course.
- You may view your selected answers only once immediately after submission before the due date. However, please note that you will not receive any score until after the due date.
- **It is your responsibility to ensure that you make the correct submission.** Failure to submit correctly does not warrant another opportunity to submit. If you choose to wait to start or submit the problem set close to the due date, you are responsible for any ensuing technical or other difficulties. You may get a grade of 0 if you accidentally start the problem set or are logged out of Blackboard, or if any of your attempts is still in progress. If you complete the problem set and plan to start another attempt, make sure you submit it before the deadline. After you launch the problem set, if possible, do not close your browser or navigate away from it until you submit the attempt. Problem sets may not be automatically submitted when the time expires.

Attendance and Participation

We expect you to be present and in good hygienic condition at every class meeting. Exceptions happen but should be rare. Attendance may be recorded in multiple ways, such as random name calls and in-class assignments. **Missing more than three class meetings during the semester is unacceptable and may result in a grade penalty.**

Participation in this course is critical to your understanding. Not participating for the majority of the semester is unacceptable. Some extra credit opportunities may only be available to those with great participation. There are two components to your participation: (1) participation during lectures in person or via Zoom (e.g., in-class assignments, Q&A, poll, etc.); (2) Piazza participation includes but is not limited to contributing to posted problems.

Classroom Courtesy: To maintain a respectful and productive learning environment, it is essential to maintain professionalism at all times. Show respect for your fellow classmates and the instructor. Disruptive behaviors, such as persistent chatting during lectures outside of designated discussion times, can significantly distract and disrupt the learning experience for others. Such behavior is not acceptable in our learning environment. Every student who attends lectures expects and deserves an atmosphere conducive to learning. **Failure to adhere to these expectations may result in a grade penalty.**

Collaboration and Honor

When you joined W&M you took the Honor Pledge: "As a member of the William & Mary community, I pledge on my honor not to lie, cheat, or steal, either in my academic or personal life. I understand that such acts violate the Honor Code and undermine the community of trust, of which we are all stewards."

Please keep in mind that you are bound by all aspects of the College's Honor Code, to which you have pledged your support. You are expected to maintain the highest standard of honesty in your work. Academic dishonesty does not serve your education well, and we will bring instances of it to the attention of the Honor Council.

Policy on outside sources: This course is meant to emulate programming and problem solving you would do in real life, so you are permitted to use any static sources you would like. This includes, books (and ebooks), journal articles, online videos, static web pages (e.g., written tutorials, blog posts, etc.). You are NOT permitted to use sites where you submit part or all of your assignment, and other individuals solve the problems/write the code for you - this includes but is not limited to sites like Reddit and StackOverflow. It is fine to read the static posts made by other people on these sites, but receiving direct assistance from individuals on the internet is not permitted. Using sites like Chegg, where individuals post questions verbatim and receive answers, is not permitted. The goal is learning, which won't happen if someone just gives you the answer. You know the difference between using a resource and asking someone to do the work for you. Additionally, do not post your code on a public repository at any time, as this violates the principles of academic honesty. For example, avoid sharing your code on platforms like GitHub or Bitbucket, as these are accessible to the public and could be used inappropriately by others.

For problem sets: You may collaborate with **up to three** other students while working on problem sets. This means groups of a maximum size of 4. Turn in your own individual assignment which has been written by you.

For projects: You may collaborate with **up to three** other students. If you consult with each other, with the instructor, with the lab instructors or consultants, or with papers or any published material, you are still required to write code independently, which means the final written solution you turn in for the assignment must be composed by you without the help of others. This means that:

- You may collaborate with individuals *currently enrolled* in CSCI 141 at W&M, but you should not be receiving assistance from individuals not currently enrolled in the course like friends, parents, siblings, random internet strangers, etc. You may neither consult students that have taken the course previously, nor their work.
- You must strictly follow the '**empty hand policy**'. You are free to discuss concepts with fellow students, but you cannot leave a discussion meeting with any record of the discussion (hard copy or electronic). All scratch paper must be torn and thrown away, boards erased, zoom meetings not recorded, etc. The granularity of collaborative discussions must be coarse enough that two people walking away from a conversation can reasonably be **expected to produce different code**. You should not look at the written solution of another student. No person other than you, the instructor, course TAs or consultants should ever see your code or your written work. If you see another student's work or allow your work to be seen by another student, then you are **both liable for an Honor Code violation** that we will prosecute.
- The actual writing of the details of a homework solution must be an individual activity, so that each student earns an individual grade for each homework assignment.
- On the assignment write-up, you must list the students you collaborated with and reference all the sources that helped you in the assignment.

We reserve the right to employ automated tools to check for plagiarism (including self-plagiarism) in both programming and written work.

Expectations for Evaluated Work

In general, we expect anything you submit to be neatly, carefully, and professionally presented. You must adhere to the file formats that we specify. **Photographs of written work embedded in PDF documents are not acceptable.**

We will not accept anything e-mailed to any instructor or assistant for this course.

All work submitted in this course, whether in draft or final form, must be your own and must be cited appropriately.

Written work should be grammatically correct. You should cite any references that you consult, noting well any restrictions on permissible items for the assignment at hand.

Programs should be well-styled. Consistent and clear naming conventions, appropriate use of comments, and efficient design are all important attributes of quality programs. Awarded scores will reflect these attributes in addition to functionality.

Artificial Intelligence

Throughout the course, you should learn to work with modern AI tools not merely as users, but as technically grounded collaborators in testing, debugging, and design. Emphasis is placed on applying foundational programming principles to audit AI suggestions, keeping the logic and evidence behind computational solutions transparent and responsible.

All work submitted in this course, whether in draft or final form, must be your own and must be cited appropriately.

You may incorporate AI-generated content or ideas in Projects, but you must indicate your use of such tools by:
i. including a formal citation to ChatGPT, etc., and
ii providing a full log of prompts, outputs, and revisions.

You remain responsible for the validity of all material generated, noting that AI-generated content often contains falsehoods and fictional sources. I strongly recommend that you carefully check all AI-generated content before submitting it for review.

The use of AI-generated content (including but not limited to ChatGPT, etc.) is strictly prohibited in Comprehension Quizzes and Problem Sets.

Student Accessibility Services

William & Mary accommodates students with disabilities in accordance with federal laws and university policy. Any student who feels they may need an accommodation based on the impact of a learning, psychiatric, physical, or chronic health diagnosis should contact Student Accessibility Services staff at 757-221-2512 or at sas@wm.edu to determine if accommodations are warranted and to obtain an official letter of accommodation. For more information, please see www.wm.edu/sas.

Diversity & Inclusion:

William & Mary values and actively nurtures an environment of diversity and inclusiveness where every individual, regardless of how we may differ – for example, but not limited to, with regard to race, religion, gender, ethnic origin, age, socioeconomic status, political preferences, physical abilities, sexual identity, or sexual orientation – is embraced, respected, and afforded the same opportunity to grow, to succeed, and to contribute to William & Mary's success.

Writing and Communication Center

The Writing and Communication Center, located on the first floor of Swem Library, is a free service provided to W&M students. Trained consultants offer individual assistance with writing, presentation, and other communication assignments across disciplines and at any stage, from generating ideas to polishing a final product. The WCC's goal is to help you become a better writer and communicator. To make an appointment, visit the WCC webpage (www.wm.edu/wcc).

Resources for Mental and Physical Well-Being

William & Mary recognizes that students juggle different responsibilities and can face challenges that make learning difficult. There are many resources available at W&M to help students navigate emotional/psychological, physical/medical, material/accessibility concerns, including:

- The W&M Counseling Center at (757) 221-3620. Services are free and confidential.
- The W&M Health Center at (757) 221-4386.

- To seek assistance for interpersonal, academic, and wellness challenges, please contact Care Support Services at wm.edu/care (care@wm.edu).
- For a list of other resources available to students, see: 

Intellectual Property

- You cannot post materials we have created for this course on public websites, including presentations, assignments, labs, quizzes, and exams.

Posting of course material: Posting content from this course on websites such as CourseHero, Chegg, Github, and Quizlet can be considered unauthorized distribution and a violation of intellectual property rights under the Student Handbook. Posting your own work allows these sites to profit from your intellectual property. Postings on these websites are visible to others, including other students. Doing so also can facilitate cheating by others who encounter and use the solutions posted.

Module 0 Learning Objectives

- have a working installation of Jupyter notebook
- be able to open, read, edit,and execute code in Jupyter notebooks
- verify that you can generate .pdf files
- introduce yourself
- understand the policies and procedures of this course
- use AI tools for environment troubleshooting and “what does this error mean?”

Module 1 Learning Objectives

- develop skills in searching, reading, and using Python documentation
- be able to identify the difference between syntax and algorithms
- understand assignment of variable names to objects and the differences between data types
- design and implement short programs using the input and print functions, and numerical and index operations
- have experience analyzing errors in code and debugging
- increase your comfort with programming in Jupyter notebooks and using a text editor
- use AI tools to critique algorithm sketches; final code must remain within the guidelines and boundaries of our course material.

Module 2 Learning Objectives

- design reusable, testable suites of Python code using functions
- import and incorporate functions from Python modules for problem-solving in program design
- generate decision-making processes in programs using conditional statements
- carry out repeated actions in programs using loop structures

- create, test, and debug increasingly complex programs using functions, conditionals, and loops
- use AI tools to design and run test cases, including edge cases, to verify that programs meet stated specifications
- critically evaluate AI-generated tests and suggestions, recognizing common AI strengths and weaknesses

Module 3 Learning Objectives

- use string and list methods to process text and generate analyzable units
- perform file input and output operations for text files
- compare and contrast lists and sets and when it is appropriate to use each
- understand the dictionary object type in general, and in the context of specific applications
- analyze text composition and contextual patterns using object methods
- translate a vague or multi-part requirement into a precise, testable specification, and use that precision to prompt AI tools effectively
- use AI as a reference when exploring unfamiliar string, list, set, and dictionary methods, and verify its explanations using official documentation and independent testing

Module 4 Learning Objectives

- understand the structure of a data frame and its utility for storing data for analysis
- create pandas dataframes from comma and tab delimited text files
- perform quality control steps on data using pandas objects and methods
- create effective and meaningful visualizations from pandas data frame using the seaborn and matplotlib modules
- as more concepts are introduced, generate and compare student-developed solutions and solutions generated by different AI tools based on correctness,

complexity, and readability; select a solution that meets course requirements and justify the choice through critical evaluation and testing

Module 5 Learning Objectives

- explore fundamental concepts of object-oriented programming
- create custom object types in Python
- write programs which make use of built-in, imported, and student-designed object types
- recognize differences between private and public attributes and when it is appropriate to use them
- use AI tools to brainstorm object-oriented designs for a given problem, including appropriate classes, attributes, constructors, and methods, while maintaining ownership of architectural decisions
- critically evaluate AI-generated class designs for alignment with course concepts